

EFT.

mse - march 2020 solution.

Q.1 $P(r, \theta, \phi) = (6, 45^\circ, 60^\circ)$.

(a)

To cartesian :-

$$x = r \sin \theta \cos \phi = \frac{3\sqrt{2}}{2} \quad \text{--- (01)}$$

$$y = r \sin \theta \sin \phi = \frac{3\sqrt{6}}{2} \quad \text{--- (01)}$$

$$z = r \cos \theta = 3\sqrt{2} \quad \text{--- (1/2)}$$

To cylindrical :-

$$\rho = r \sin \theta = 3\sqrt{2} \quad \text{--- (01)}$$

$$\phi = \phi = 60^\circ \quad \text{--- (1/2)}$$

$$z = r \cos \theta = 3\sqrt{2} \quad \text{--- (01)}$$

Q.1(b) $\phi = xyz$

$$\text{Gradient}(\phi) = \nabla \phi$$

$$F = \frac{\partial \phi}{\partial x} \bar{i} + \frac{\partial \phi}{\partial y} \bar{j} + \frac{\partial \phi}{\partial z} \bar{k} \quad \text{--- (02)}$$

$$= yz \bar{i} + xz \bar{j} + xy \bar{k}$$

To prove F is irrotational

$$\nabla \times F = 0$$

$$\nabla \times F = \begin{vmatrix} \bar{i} & \bar{j} & \bar{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ F_x & F_y & F_z \end{vmatrix} = 0 \quad \text{--- (03)}$$

Hence irrotational.

Q.1(c) Statement ——— (02)

Explanation. ——— (03)

Q.2. $\vec{D} = 5r^2 \hat{a}_r \quad r < 0.08 \text{ m.}$
 $\vec{D} = \frac{0.1}{r^2} \hat{a}_r \quad r > 0.08 \text{ m.}$

$$\rho_v = \nabla \cdot \vec{D}$$

$$= \frac{1}{r^2} \frac{d}{dr} (r^2 D_r)$$

$$= \frac{1}{r^2} \frac{d}{dr} \left(r^2 \frac{0.1}{r^2} \right) = 0$$

$$\boxed{\rho_v \Big|_{r=0.1} = 0}$$

————— (04)

$$\rho_s = \frac{Q}{4\pi r^2}$$

$$Q = \int_V \rho_v \cdot dV = \int_0^{2\pi} \int_0^\pi \int_0^{0.08} \rho_v \cdot dV \quad \text{————— (01)}$$

$$dV = r^2 \sin\theta \, dr \, d\theta \, d\phi$$

————— (01)

$$\rho_v = \nabla \cdot \vec{D} = \frac{1}{r^2} \frac{d}{dr} (r^2 D_r)$$

$$= \frac{1}{r^2} \frac{d}{dr} [r^2 \cdot 5r^2]$$

$$= \frac{1}{r^2} \cdot 20r^3 = 20r \text{ C/m}^2$$

————— (02)

$$Q = \int_0^{2\pi} \int_0^\pi \int_0^{0.08} 20r \cdot r^2 \sin\theta \, dr \, d\theta \, d\phi$$

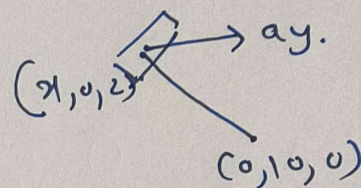
$$Q = 20 \times 10^{-3} \left[\frac{r^4}{4} \right]_0^{0.08} [-\cos \phi]_0^{\pi} [\phi]_0^{2\pi}$$

$$Q = -2.57 \mu C.$$

$$\rho_s = - \frac{2.57 \times 10^{-6}}{4\pi (0.08)^2} = -32 \mu C/m^2$$

Q.3 $\vec{K} = 20 \vec{a}_x \text{ A/m}$

$$dH = \frac{K \cdot dS \times \vec{a}_R}{4\pi R^2}$$



dS at point $(x, 0, z)$ on $y=0$ plane.

$$\vec{R} = -x\vec{a}_x + 10\vec{a}_y - z\vec{a}_z$$

$$\vec{a}_R = \frac{-x\vec{a}_x + 10\vec{a}_y - z\vec{a}_z}{\sqrt{x^2 + 100 + z^2}}$$

$$dS = dx \cdot dz \vec{a}_y$$

$$dH = \frac{20\vec{a}_x dx \cdot dz}{4\pi} \left[\frac{(-x\vec{a}_x + 10\vec{a}_y - z\vec{a}_z)}{\sqrt{x^2 + 100 + z^2}} \right] \cdot \vec{a}_y \cdot \sqrt{x^2 + 100 + z^2}$$

$$= \frac{5(10\vec{a}_z + z\vec{a}_y)}{\pi(x^2 + 100 + z^2)^{3/2}} dx \cdot dz$$

$$H = \int dH = \frac{5}{\pi} \left[\int_{-\infty}^{\infty} \int_{-5}^5 \frac{10\vec{a}_z dx \cdot dz}{(x^2 + z^2 + 100)^{3/2}} + \int_{-\infty}^{\infty} \int_{-5}^5 \frac{z\vec{a}_y dx \cdot dz}{(x^2 + z^2 + 100)^{3/2}} \right]$$

$$= \frac{5}{\pi} \left[\int_{-5}^5 \frac{20a_2 dz}{z^2+100} + \int_{-5}^5 \frac{22a_y dz}{z^2+100} \right]$$

$$= \frac{5}{\pi} \left[\frac{20}{10} a_2 \left[\tan^{-1} \left(\frac{z}{10} \right) \right]_{-5}^5 + a_y \left[\ln(z^2+100) \right]_{-5}^5 \right]$$

$$\overline{H} = 2.952 \overline{a_2} \text{ A/m}$$

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